What's the Difference Between Malware, Trojan, Virus, and Worm etc…

One of the most frequently (and painfully) misused groups is the terms that differentiate malware from other types of vulnerabilities and threats. Malware, trojans, viruses, and worms are and how they're different from one another.

A computer virus is a type of malware. Malware includes computer viruses, worms, Trojan horses, spyware, ransomware and many others. Wannacry is a form of ransomware.

- Malware is the shortened form of **Malicious software**.
- Malware is an umbrella term used to refer to a variety of forms of hostile or intrusive software including Ransom wares, Computer Viruses, Worms, Trojan Horses, Spyware, Adware, Scareware etc.
- This is any program or file that is harmful to a computer user.
- The term refers to software that is deployed with malicious intent.
- Malware can be deployed even remotely, and tracking the source of malware is hard.
- It can take the form of executable code, scripts, active content, and other software.

Malware is typically used:

- To steal information that can be readily monetized, such as login credentials, credit card and bank account numbers,
- And intellectual property such as computer software, financial algorithms, and trade secrets.
- To ransom money in Bitcoin, for example, Wannacry Ransomware.
- Spy on computer users for an extended period without their knowledge, for example, Reign Malware.
- Extort payment for example Cryptolocker.

List of Common Malware types:

- **Adware**: The least dangerous and most lucrative Malware. Adware displays ads on your computer.
- **Spyware**: Spyware is software that spies on you, tracking your internet activities in order to send advertising (Adware) back to your system.
- **Virus:** A virus is a contagious program or code that attaches itself to another piece of software, and then reproduces itself when that software is run. Most often this is spread by sharing software or files between computers.
- **Spam:** Spamming is a method of flooding the Internet with copies of the same message. Most spams are commercial advertisements which are sent as an unwanted email to users. Spams are also known as Electronic junk emails or junk newsgroup postings. These spam emails are very annoying as they keep coming every day and keep your mailbox full.
- **Worm:** A program that replicates itself and destroys data and files on the computer. Worms work to “eat” the system operating files and data files until the drive is empty.
- **Trojan:** A Trojan horse or Trojan is a type of malware that is often disguised as legitimate software. Trojans are written with the purpose of discovering your financial information, taking over your computer’s system resources, and in larger systems creating a “denial-of-service attack” which is making a machine or network resource unavailable to those attempting to reach it. Example: Google, AOL, Yahoo or your business network becoming unavailable.
- **Ransomware**: If you see this screen that warns you that you have been locked out of your computer until you pay for your cybercrimes. Your system is severely infected with a form of Malware called Ransomware. Even if you pay to unlock the system, the system is unlocked, but you are not free of it locking you out again.
- **Backdoors:** Backdoors are much the same as Trojans or worms, except that they open a “backdoor” on a computer, providing a network connection for hackers or other Malware to enter or for viruses or SPAM to be sent.

How do computer viruses work?

To understand how computer viruses work, it's helpful to split them into two categories: those that begin to infect and replicate as soon as they land on your computer, and those that lie dormant, waiting for you to unwittingly execute the code.

Computer viruses have four phases (inspired by biologists' classification of a real-life virus's life cycle).

- **Dormant phase:** This is when the virus is hidden on your system, lying in wait.
- **Propagation phase:** This is the *viral* stage, when the virus begins to self-replicate, stashing copies of itself in files, programs, or other parts of your disk. The clones may be slightly altered in an attempt to avoid detection, and these copies will also self-replicate, creating more clones that continue to copy and spread.
- **Triggering phase:** A specific action is generally required to trigger or activate the virus. This could be a user action, like clicking an icon or opening an app. Other viruses are programmed to come to life after a certain amount of time, such as a logic bomb designed to trigger after your computer has rebooted a certain number of times (this is done to obfuscate the origin of the virus).
- **Execution phase:** Now the virus's program is executed and releases its **payload**, the malicious code that harms your device.

Different Hacking Methods

Phishing

Phishing is a form of fraud in which an attacker masquerades as a reputable entity or person in email or other forms of communication. Attackers will commonly use phishing emails to distribute malicious links or attachments that can perform a variety of functions. Some will extract login credentials or account information from victims.

Phishing attacks typically rely on social networking techniques applied to email or other electronic communication methods. Some methods include direct messages sent over social networks and SMS text messages.

Phishers can use public sources of information to gather background information about the victim's personal and work history, interests and activities. Typically through social networks like LinkedIn, Facebook and Twitter. These sources are normally used to uncover information such as names, job titles and email addresses of potential victims. This information can then be used to craft a believable email.

Typically, a victim receives a message that appears to have been sent by a known contact or organization. The attack is then carried out either through a malicious file attachment, or through links connecting to malicious websites. In either case, the objective is to install malware on the user's device or direct the victim to a fake website. Fake websites are set up to trick victims into divulging personal and financial information, such as passwords, account IDs or credit card details.

D-DOS Attack

A DDoS, or distributed denial-of-service, attack is an attempt by a cybercriminal to flood a server with traffic to overwhelm its infrastructure. This causes a site to slow to a crawl or even crash so legitimate traffic won’t be able to reach the site. This type of attack can do a lot of damage to your online business.

Most DDoS attacks are done with botnets – groups of computers all acting together. These computers will all attempt to access a website simultaneously, overwhelming the server and bringing it down.

How do they get these botnets? By hijacking other machines. Often, a hacker will use malware or take advantage of an unpatched vulnerability on someone else's server to gain access to it via Command and Control (C2) software. By leveraging these exploits, hackers are able to amass large numbers of computers in a relatively cheap and easy way, which they can then deploy for their own nefarious purposes.

Once they have control over enough machines, the hackers can then issue a command to the entire botnet, which then attempts to access the target server. When too many computers are trying to access a server all at once, service outages are common. The end result is an interruption in service and lost productivity.

SQL Injection

An SQL injection attack consists of an insertion or injection of a SQL query via the input data from the client to the application. SQL commands are injected into data-plane input that affect the execution of predefined SQL commands. A successful SQL injection exploit can read sensitive data from the database, modify database data (viz., insert, update, or delete), execute administrative operations on the database, recover the content of a file present in the database management system, and even issue commands to the operating system in some instances.

To perform an SQL injection attack, an attacker must locate a vulnerable input in a web application or webpage. When an application or webpage contains a SQL injection vulnerability, it uses user input in the form of an SQL query directly. The hacker can execute a specifically crafted SQL command as a malicious cyber intrusion. Then, leveraging malicious code, a hacker can acquire a response that provides a clear idea about the database construction and thereby access to all the information in the database.